PHY201: Waves & Optics

Problem Set 7

November 2, 2025

1. Two strings of tension T and mass densities μ_1 and μ_2 are connected together. Consider a travelling wave incident on the boundary. Find the ratio of the reflected amplitude to the incident amplitude, and the ratio of the transmitted amplitude to the incident amplitude, for the cases $\mu_2/\mu_1 = 0$, 0.25, 1, 4, ∞.

2. Two strings of tension T and mass densities μ_1 and μ_2 are connected together. Consider a travelling wave incident on the boundary. Show that the energy flux of the reflected wave plus the energy flux of the transmitted wave equals the incident energy flux (energy flux of a wave is defined as the product of energy density and wave velocity).

3. If the electric field in free space is $E = E_0(\hat{\mathbf{x}} + \hat{\mathbf{y}}) \sin[(2\pi/\lambda)(z + ct)]$ with $E_0 = 20$ volts/m, then the magnetic field, not including any static magnetic field, must be what?

4. Write out formulas for E and B that specify a plane electromagnetic sinusoidal wave with the following characteristics. The wave is traveling in the direction $-\hat{\mathbf{x}}$; its frequency is 100 megahertz (MHz), or 108 cycles per second; the electric field is perpendicular to the $\hat{\mathbf{z}}$ direction.

5. Here is a particular electromagnetic field in free space :

$$E_x = 0 \quad E_y = E_0 \sin(kx + \omega t) \quad E_z = 0$$
$$B_x = 0 \quad B_y = 0 \quad B_z = -\frac{E_0}{c} \sin(kx + \omega t).$$

Show that this field can satisfy Maxwell's equations if ω and k are related in a certain way.

6. Show that the electromagnetic field described by

$$\begin{aligned} \mathbf{E} &= E_0 \hat{\mathbf{z}} \cos kx \cos ky \cos \omega t \\ \mathbf{B} &= B_0(\hat{\mathbf{x}} \cos kx \sin ky - \hat{\mathbf{y}} \sin kx \cos ky) \sin \omega t \end{aligned}$$

will satisfy Maxwell's equations in vacuum if $E_0 = \sqrt{2} c B_0$ and $\omega = \sqrt{2} ck$.

7. A light pulse from a ruby laser consists of a linearly polarised wave train of constant amplitude lasting 10^{-4} s and carrying energy of 0.3 J. The diameter of the circular cross-section of the beam is 5×10^{-3} m. Calculate the energy density in the beam and find the root mean square value of the electric field in the wave.